Chatter and Listening: A wireless multi-node sound sculpture

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Abstract

'Chatter and Listening' is a project that creates a number of re-locatable art objects with sonic attributes. These objects act as single entities; they can create decisions, observe their environment and operate alone. They also act in the narrative of a colony where each entity is guided in behaviour by the state of the colony and its surrounding environment. The project aims to create a multi-node spatial sound sculpture that operates wirelessly over a range of distances to create a distributed sonic art object.

'Chatter and Listening' uses technologies including embedded micro controller systems and wireless communications combined with other artistic practices to realise an intermedia artwork.

This paper describes the concepts, technologies and a composition philosophy behind the work. It also looks at the behavioural model of the Australian Magpie that is used in creating objects that communicate and generate sound over a wide spatial location.

Introduction

Sound sculpture is the combination of sonic and sculptural art practices. It draws stimuli from fields that include acoustic design, architecture, electronics, design and fabrication of physical objects, intermedia practices, music and sound design. The output of this practice can take many different forms ranging from auditory and physical embodiments to simulations inside a computer system. Sound sculpture re-defines itself with every work and re-codifies the space in which it resides.

Acoustic attributes that are inherent to sound sculpture can express narratives and abstractions that transmute the physical intersections of the object with the space it inhabits. This paper describes a sounding work from the concept up. (The concepts are important to establish as they define the direction that the implementations of the physical materialization, wireless communications and the sonic composition are to take.)

'Chatter and Listening' explores narratives in a colony of sounding objects by employing the behavioral model of a natural system. To date the majority of musical and sound compositions that utilize behavioural models of living dynamic systems have comprised of a sonification of single agent movements in space through time in the meta-framework of a colony. Common models employed include schools of fish and flocks of birds. These works produce a composition when a number of the agents interact through a set of defined behaviours, sonifying the school or flocks positioning in space through time (Blackwell 2002).

Other dynamic systems are used in music/sound creation. These include evolutionary algorithms that use a breeding and selection criteria to produce musical information (Todd 1999). A number of agents are encouraged to interact by breeding with other agents that are deemed fit by a set of programmed criteria. The breeding produces a number of offspring that carry traits of both parents by genetic cross over, and mutations are introduced to liven up the genetic pool.

The artificial ecosystem (Conrad 1985) is an environment where virtual organisms move about interacting with each other and the changing environment they inhabit. As in the real world this environment and its inhabitants change dynamically over time. For example, depending on the time of the day or generation behavioural changes can be seen; different generations express individual characteristics (traits) evolving from the parents; roles of behaviour change from predator to prey. The result of this implementation in a sonic artwork is a sonification of the organisms state parameters into an audible work that provides intrinsic results (Dorin 2006).

One undertaking of this project is to map a behavioural model of a *single* species of animal to provide a successful composition paradigm for a sound space. The exhibited behaviours will be analogous to the real world interactions that the animal displays. Stimulations of the inhabited environment, other individuals in the group and other species all provide influence to the behaviours. These behaviours are employed are implemented at a program level on embedded technologies which are implanted in the distributed sculptural nodes.

The interaction the colony nodes have with one another is two fold. Communications of an individual state and intentions (behaviours), and the state of the colony at large, is carried out on a 2.4GHz wireless broadband network. Sonic materializations of the individual and colony at time are actuated by the behaviours and communications. The sonic events can be received, depending on an individual's location, and realised as another communication expression.

Related works

Electronic Sonic

Early use of electronics to create sonic output in sound sculpture can be seen with Percy Grainger's 1952 work *'The kangaroo Pouch Free Music Machine'*. In this work were eight oscillators that were controlled by a largish sheet of brown paper with pitch control graphs in different colours (score) that was fed from one roller (the "Feeder") to another (the "Eater") driven by a motor. The sonic output form this machine materializes as a wild sweep of the oscillators gliding over the intervals set out on the graphs.

Sounding Objects

The form of an object can facilitate the perception of a life force. This can be seen with sculptural practices that aim to represent the embodiment of living forms, either in traditional or conceptual narratives as in the work of Patricia Piccinini and her project *'we are family'* 2003[1]. The essence of a living thing is inherently multi dimensional; form being one dimension, so projecting different or multiple dimensions on an object has the potential to give varying perceptions of a life force in that object.

'*Cyber Squeeks'*, Ken Rinaldo 1994, imitates intelligence of machine life forms by creating a number of organic looking creatures that create sound based on input from various activations, for example viewer interaction. The response of the objects is a tuneless welter, signifying a life like response to stimulation.

Location location

When observing or interacting with a sounding object the viewer is engaged in a multi-sensory experience, that requires an amount of listening as well as viewing the physical embodiment. The innate physicality of sculpture allows it to be viewed from many angles and the overall observation on the object and space is sometimes an important contributor to the work.

A sound work (since the advent of tape fetishism) is viewed primarily in a two-dimensional sound field i.e. Left, Right. However in discussing the qualities of a sculptural object the listening of the sound field emanating takes on a multifarious nature, where the viewers position dictates the immediate perception of how and when it is heard redefining the work in space.

The work '*finitorium' 1993*, by Chris Ulbrick explores the subjectivity of location in relation to an object and it's spatial situation. Sound is mapped in the installation space by positional information of the viewers at time that is relayed to a computer by sensors. The effect is of changing the spatial relationship of the voice and abstract audio elements in the vertical and horizontal plane, which is different every time because the number of people viewing the work and their distribution will always change.[2]

Concepts

In this section a number of key concepts will be defined that will relate to the dimensions of the project at an abstract level. The ideas that these concepts bring to the table will underpin the directions of the other components in this paper.

Natural Systems

Individual co-located agents of species of organism can be seen in a host of naturally occurring co-operative systems as colonies. A colony is a complex system of individuals interacting with each other and their environment (Booker, 2004). Colonies have a hierarchy with at least one order of individual higher than another and they appear to act co-operatively and purposefully.

Colony behaviour emerges from auditory cues not necessarily within our range of hearing, visual displays, contact between individuals and environmental pressures.

Cognitive perception

When functioning correctly individuals have an awareness of the representation of the environment that they exist within. This is a multimodal experience built from unimodal cortical responses including visual, auditory and somatosensory association. A perceptual/cognitive system extracts dimensional information from the stimulus array (Lawrence, 1981). If there is a change for some reason in the environment then those stimuli will be drawn to our attention and our perception of the multimodal locality will adapt. (Downar, 2002)

Core Understanding

Behaviours of individuals in a reasoning society arise from streams of information from the immediate environment, and perceptions in the state of the self in relation to others in the group. When same society individuals communicate there is an empathy that is constructed from historical memory and the perception of a future self in relation to that society.

[1] http://patriciapiccinini.net/ cited 29/03/07

[2] Some observations of the works in section 2 were derived from Ross Bandt's book and accompanying CD Sound Sculpture: Intersections In Sound & Sculpture In Australian Artworks

Behavior as Composition

On a theoretical level *'Chatter and Listening'* looks at using the behavioral model of living systems for electroacoustic music composition. This paper argues that musical and sounding works are based on different instigators of behaviours. An example of this is with a scored composition, where an instruction set telling a performer how to behave with their instrument produces the desired outcome of a composer. In sound designs events are more directly related to occurring behaviours of real world materializations, where the time and space of the sound event is placed according to some acknowledged auditory 'what' and 'where' subsystem (Kubovy 2000). An example of this is with a familiar bird vocalising in a tree. You are familiar with the sounds the creature makes and you know of its relative position from visuomotor responses ('where'). Your auditory system acknowledges the edges of the sound, where the frequencies and amplitudes begin and end 'what', defining an auditory object. When a number of these objects develop in structure and interact based on dynamic and purposeful behaviours the product displays new meanings.

Adaptive Behaviour

In the article Designing and Understanding Adaptive Group Behavior Maja (1995) introduces the concept of 'basis behaviors' for the building blocks of synthesized group behaviors. In *'Chatter and Listening'* basis behaviors are used to operate each nodes behaviors, thereby forming the composition. A set of basis behaviors has criteria that make for maximum efficiency based on relevance to obtaining goals or helping obtain desired goals and how each behavior relates to another. Each basis behavior cannot be achieved with any other of the other basis behavior in that set or reduced to them. Maja (1995) gave autonomous control to 20 mobile robots using basis behaviors. With the implemented basis behaviors adaptive agent control, social interaction and learning was achieved in the robots.

When designing basis behaviors a number of techniques emerge from the article for finding an effective and efficient set. One suggested approach used in *'Chatter and Listening'* includes: firstly finding a naturally occurring organism with a set of behaviors that may appeal to the wanted outcomes as a good starting point for finding a package of relevant and efficient base behaviors. This can be the case because the species to be looked at has refined the behaviors through many generations of evolution for maximum gain and efficiency. Also of use may be putting a list of behaviors into an enumerated table and evaluating them on the relevance they have to the desired goals and the relation they have to one another. From this process a number of quality base behaviors should emerge.

Behaviours of Natural Living Systems

Various species that operate in natural dynamic systems including wasps, ants, bees and birds have been observed and their behaviours mapped out. This mapping has given researchers new ways of approaching and efficiently managing modern complex systems. Although the individuals operate with an amount of simplicity the colony structure provides a distributed and complex system that is capable of definite tasks, which exceed the capabilities of a single individual.

The observation of birds in nature is a fascinating insight into the behaviours of living complex systems. As a higher-level approach *'Chatter and Listening'* includes the practice of natural system observations. Craig Reynolds observed the activities of fish and birds in his seminal work on flocking simulations (Reynolds C. 1987). Flocking, in a computer science term, is a simulation of group behavior primarily associated with the visual domain; examples include a flock of birds or a school of fish. It is a type of optimization in creating complex systems of sociological/biological behaviors. A number of sound based works have used these models based on dynamic natural systems as a type of musical score (Klein, 2002) (Spector 2003). The movements of the individuals or the flock in space are mapped to different musical pitches and when put into the context of the flock over a time it will produce a complex and evolving piece of music, a sonification of the flock.

Species Australian Magpie

The Australian Magpie is a medium sized black and white songbird, found across the Australian continent. There are several subspecies scattered across the country, but there are definite behavioural characteristics that they have in common. The species behavioural model has been chosen for the project because of the personal affinity and positive observations the author has had for the animal. Since European settlement in Australia the Magpie has shown to be highly adaptable and sociable, where they are able to sustain populations with the encroachment of human development, often interacting with people.

Flocks of Magpies tend to be no more than seven at one time, and these small groups will most often be seen using their skills to hunt at ground level, taking to the higher levels of the trees only for roosting and breeding. When feeding Magpies use a method of extraction foraging that involves location of prey by listening for the sound of vibrations made by potential food makes and not with visual cues (Simpson 1993). After extraction the Magpie will often hide and later retrieve food in a complex manner using a memory skill called caching.

Magpies have a high level of individuality and have some unique behaviour such as social play and cooperative behaviour with same and different species. Below (Table 1) is a 'basis behaviour' table (see beginning of section 5) that outlines the appeal the Magpie exhibits for behaviour set implementation.[1]

Behaviour name	Behaviour function	Skills
Flocking	Safety in numbers Social permutations	Entity Recognition.
Extraction Foraging	Food gathering	Listening. Location identification.
Caching	Resource control	Memory
Cohesion	Effective teamwork	Entity Recognition. Task Reasoning. Communication.
Agonistic	Nest protection	Aggressive response. Vocalisation.
Social Play	Learning to handle prey Social bonding Education in food, risks, predators, vocalisation	Pseudo aggressive response. Communication. Vocalisation.
Alarm calling	Flock protection	Entity Recognition. Vocalisation.
Courtship	Female calls to male for procreation	Sexual awareness. Vocalisation.

Table 1. *Basis behavior table of the Austalian Magpie*

Communications

Each of the sounding objects in *a 'Chatter and Listening'* colony makes decisions based on the behavior criteria, a number of activities then occur on the 2.4GHz data band and the audio level. A telemetry device contained within the vessel transmits and receives communication of machines therefore relaying their behavior. This data communication is the means that all the machines in the colony use to configure the dynamics of the system at any given time.

The sounding element of the work is the auditory embodiment of the behaviours programmed into the colony. Just as a natural living system that communicates with a sonic language has an impact to the external world the colony will convey itself at an audio rate.

Songbirds

Animals communicate in a variety of ways such as auditory communication, visual displays, touch and smell, and one of the most efficient types of communication for a bird species is through vocalisation (song). Bird song is able to be transmitted over long distances very quickly, then to be received and decoded by another of that family group or another species. This type of communication where an entity transmits and all other entities can hear is a broadcast communication. The song defines an intention and is purposeful. Two examples from the basis behaviour table that are different but both requiring vocalisation are alarm calling and social play. In alarm calling one magpie will view a threat and make a decision to raise an alarm start producing the definite vocal signature that alerts the other magpies of that family to take action. This is unlike in a social play situation where two or more magpies will start playfully squabbling with each other, in a physical sense. The vocalisations are to acknowledge and demonstrate that the current activity is not aggressive.

Wireless Sensor Networks

Development of technologies and ontology's in wireless sensor networks has created advancements in this field recently. These networks are deployed for various applications like environmental monitoring of microclimates, patient and drug monitoring in hospitals and environment control in offices (Akyildiz 2002).

A network of this type comprises of *n* number of wireless sensor nodes in an environment, not placed at any definite position, and because of this the means by which the connections the nodes have to relay any piece of information from *a* to *b* becomes important, possibly complex as nodes may drop in and out of a network for a number of reasons at any given time. The way in which the connections are configured for communication is called the network topology.

Wireless peer to peer sensor networks use a topology, generally a Mesh topology (see figure 1), that encourages low transmission distances and more efficient network communication by allowing nodes to only communicate with their nearest neighbour[2]. In this type of wireless network topology each node is considered the same as the other, and with these networks distributed over possibly large distances messages are relayed with an understanding that each node will route another's message. The Mesh allows distributed nodes to be

[1] Sets of the behaviour of the Australian Magpies in section 5.2 were derived from "Australian Magpie, Biological and Behaviour of an Unusual Songbird", Gisela Kaplan 2004

[2] A summary on Mesh topologies can be viewed at http://en.wikipedia.org/wiki/Wireless_mesh_network. cited 19/03/2007

connected along a number of different paths to route data, and because of this the communications link is self-configurable as nodes drop out or new ones join in. (Akyildiz 2005)

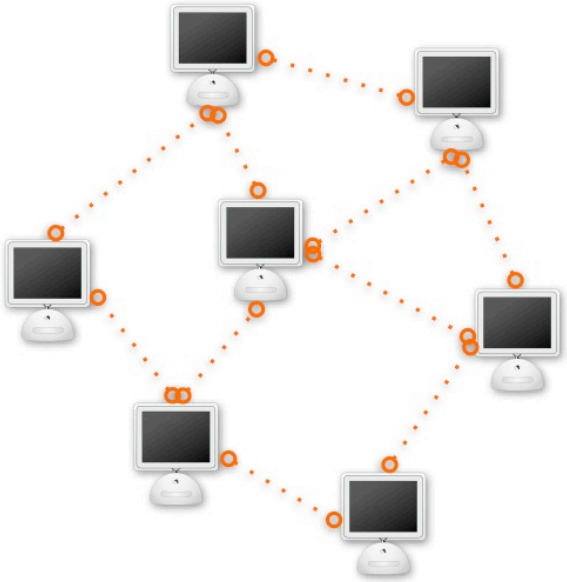


Figure 1. *Visualization of a Mesh peer-to-peer network.*

Hardware technologies

Each machine is based on a modular design, which allows reconfiguration for installations and less expense in replacement if only single module goes bust in the system. The modules that are at the core of the system are:

- micro controller unit
- wireless transceiver
- MP3 player/recorder
- Sensor array

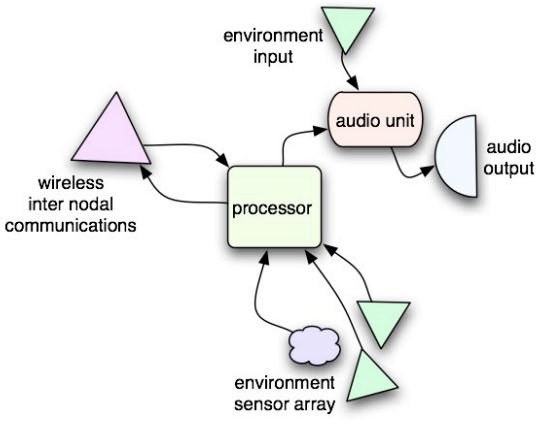


Figure 2. *Hardware technologies abstract diagram.*

The wireless communications requires a transceiver that has the ability to communicate state information of the unit and other units in real time. For this a MaxStream XBee Pro™ module that transmits at 2.4GHz with 60mW power [1]. This is a modern device that supports complex topologies, addressing and error handling. A sensor array relays the condition of the environment to the unit and colony on demand and consists of microphones to relay sound pressure levels, a light dependent resistor to gauge the light level of the environment and a temperature sensor.

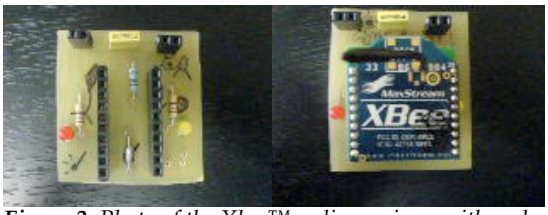


Figure 3. *Photo of the Xbee™ radio carriage, with and without the module installed.*

The audio unit defines the sonic attributes of the machine and is able to randomly access a database of sound designs to be played back. Recording functionality is also included to take segments of the sonic environment to play back at a latter time. A number of lower level MP3 units are available from different suppliers and have become more prevalent since the introduction of low cost encoder/decoder IC technologies. The lower level MP3 units that were researched did not fulfil all the necessary requirements, i.e. they were either too expensive or did not have all the desired features.

To overcome this an off the shelf MP3 player/recorder was chosen. It has the requirements and is also simple enough electromechanically to install a new switching mechanism, and in functionality to manipulate electronically by the micro controller.

Firstly the buttons are removed and hook-up wire attached to the necessary pads.



Figure 4. *Photo of the opened MP3 unit with hook up wire connected to the button pads.*

Then a transistor array is used as a switching block for all the controls.

[1] Information for the XBee Pro from MaxStream can be found at http://www.maxstream.net/products/xbee/xbee-pro-oem-rf-module-zigbee.php. cited 19/03/2007

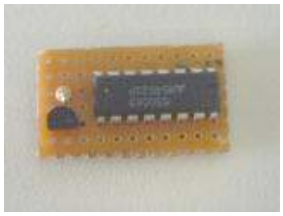


Figure 5. *Photo of the transistor array used as the new switching mechanism. The small transistor/resistor pair on the left is for the play/stop function as it runs on a different power rail.*

And finally all the wires and new circuit board are loaded into the back of the unit and the case returned.

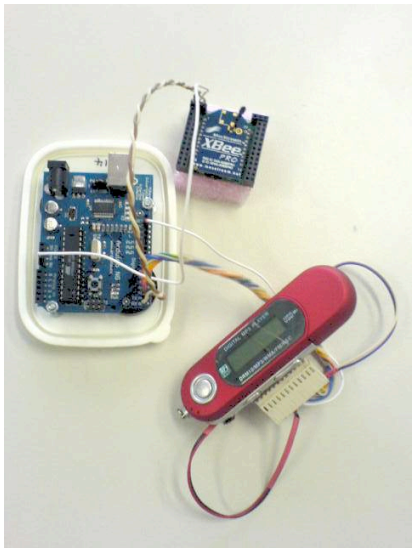


Figure 6. *Photo of the core components for the hardware system. Note the new connector block on the bottom of the MP3 player/recorder.*

Of course a speaker and embedded amplifier will also be required to externalise the audio. The micro controller performs all the calculations from the sensors and wireless input to make decisions on the state of the entity and what sonic attributes it will have at time.

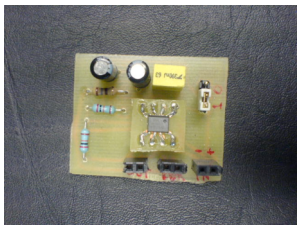


Figure 7. *Photo of the 3W LM4991MA amplifier circuit.*

Software abstractions

The implemented behavior systems and hardware technologies are at the service of a type of awareness the nodes will process. A node will have awareness of space through the environment sensors relaying information about the space they inhabit. An awareness of the group and their place within the group will exist by the data level information telemetries. This awareness is configured and realized by the software.

Creating self-awareness in communication nodes becomes possible when the data they are transferring as well as their own and other nodes capabilities are 'understood' (Wang 2003). The capabilities that each node possesses for purposes of this project are the same as any other node that will be communicated with. A decision unit from time to time will request a group check, which sends a request from the radio that all nodes in the group are inclined to respond. The responses are cross-referenced with an array of the last known members in the group; if there are any differences then the array is updated. This means if a node drops out or is no longer able to communicate then the topology reconfigures itself and capability of all nodes is known as good.

To make sense of the data received from the radio and environment sensors, an understanding unit distributes the data to the destination requesting it or passes it to the decision unit to process. For example let's say the temperature rises for some reason, referring to figure 8, the environment monitor notices a difference and compares it to a database of the past temperatures, finding the difference then requesting the understanding unit to do something with the data. The understanding unit takes the difference and (in this case) interprets it as hotter, understanding then requests the decision unit to do something. The decision unit will then request the state of the machine to update accordingly and radio the ID of itself and the new state to the rest of the colony, unless for some reason it decides not to.

In each behavioral state there is a set of sonic attributes that relate to that state. The state is set by the decision unit and is based on streams of information from the immediate environment and the state of the self in relation to other nodes in the colony. Separate from the set state is the state dynamics unit that takes input from all meaningful sources to create dynamics to the sonic motif. Requests are made to gather information on the current state of the machine, environmental data, the state of other colony members and request are made to other nodes in the colony to understand what their particular state dynamics may be. An example of this latter point could be that the state dynamics unit wants to create a harmonic relation within the colony after checking the state of other colony nodes and environmental data. A request would be sent out the radio to the state dynamics unit in other nodes in the colony to exhibit a part of a harmony. The nodes would then respond if they would comply or not. This unit communicates with the MP3 player/recorder and commands it's functions.

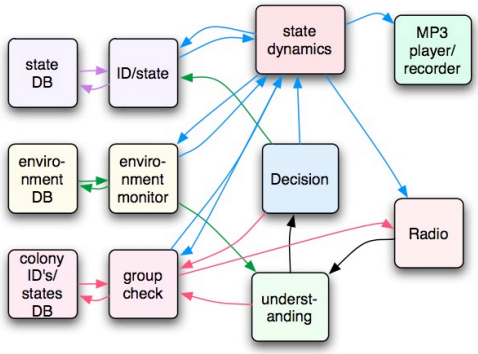


Figure 8. *Ontology of software abstraction examples.*

Conclusion and future work

Using a behavioural model for a composition paradigm embedded in a colony of sculptural forms leads to diverse applications for use in a distributed art object. This paper has mapped out a range of concepts and a theoretical realization of these through dynamic systems of behaviour. Furthermore it describes the Australian Magpie as a single behavioural model to adopt for the use in a sound sculpture work.

The means in which objects exhibiting the adopted behavioural model are able to communicate leads to exciting avenues in wireless communications topologies on the level of wireless sensor networks. An application of a wireless topology as well as the hardware and software ontologys of the theoretical components has been explained for the use in a distributed artwork.

The full realization of a body of work that adopts the processes outlined in this paper is to be carried out. The final work is to be produced for gallery exhibition and installation in space. A robust package is being developed that will cope with the handling and other stresses of these types of environments.

Once a package has been designed then it will be portable to other projects and other artists to use. The range as well as the type of communication and the reconfigurable nature of the technology means that the scale of the project that implements what is outlined in this paper has the ability to encompass an area the size of the corner of a room to networking over the area of a city.

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References

Akyildiz, I. Su, W. Sankarasubramaniam, Y and Cayirci, E. 2002. "Wireless Sensor Networks: a survey", *Computer Networks,* vol 38:4, pp 393-422.

Akyildiz, I. Wang, X and Wang, W. 2005. "Wireless Mesh Networks: a survey". *Computer Networks,* volume 47:4, pp 445-487-422.

Bandt, Ros. 2001"Sound Sculpture: Intersections In Sound & Sculpture In Australian Artworks". *Craftsman House.*

Blackwell, T. and Bentley, P. 2002. "Improvised music with swarms". *Proceedings of the 2002 Congress on Evolutionary Computation 2002.* vol 2, p 1462-67.

Booker, Rebecca. 2004. "What Ants and Bees Tell Us about Social Network Theory". *AAAS Dialogue on Science, Ethics, and Religion January 2004.*

Conrad, M. and Strizich, M. 1985. "A Computational Model of an Evolving Ecosystem". *Biosystems 17.* p 245 - 258.

Dorin A. 2006. "The Sonic Artificial Ecosystem". *Proceedings of the Australasian Computer Music Conference 2006.* p 32 – 37.

Downar, Jonathan. Crawley, Adrian P. Mikulis, David J. and Davis, Karen D. 2000 "A multimodal cortical network for the detection of changes in the sensory environment". *Nature Neuroscience.* vol 3, p 277 – 283.

Kaplan, Gisela. 2004. "Australian Magpie, Biological and Behaviour of an Unusual Songbird". *CSIRO publishing.*

Kubovy M. and Valkenberg, Van. *2000* "Auditory and visual objects". *Cognition.* vol 80, p 97-126.

Lawrence M. Ward James. and Russell, A. 1981 "Cognitive Set and the Perception of Place". *Environment and Behavior.* vol 13, p 610.

Maja, Mataric. 1995 "Designing and Understanding Adaptive Group Behavior". *Adaptive Behavior.* vol 4, p 51.

Reynolds, Craig. 1987. "Flocks, Herds, and Schools: A Distributed Behavioural Model". *Computer Graphics*, vol 21:4.

Simpson, K. Day, N. and Trusler, P. 1993 "Field Guide to the Birds of Australia 'Simpson and Day'". *Penguin for Lifetime Distributors*. New South Wales, Australia.

Spector, L. Klein J. Perry, C. and Feinstein, M. 2003 "Emergence of Collective Behavior in Evolving Populations of Flying Agents" *In Proceedings of the Genetic and Evolutionary Computation Conference*. Springer-Verlag. p. 61-73.

Todd, P.M. and Werner, G.M. 1999. "Frankensteinian Methods for Evolutionary Music Composition"
Musical networks : parallel distributed percetion and performance. Cambridge, MA MIT Press. xv, 385.

Wang, J. Brady, D. Baclawski, K. Kokar, M. and L Lechowicz. 2003. "The Use of Ontologies for the Self-Awareness of the Communication Nodes". *In Proceedings of the Software Defined Radio Technical Conference*.